Xixian Niu

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EDUCATION

Ph.D. in Economics, Université Libre de Bruxelles (ECARES and I3h) — expected 2027
Supervisors: Micael Castanheira and Mathias Dewatripont

Visiting Ph.D. Student, Toulouse School of Economics — March – April 2027 (forthcoming)
TSE Health Center, host: Prof. Pierre Dubois

Visiting Researcher, Imperial College London — May 2023 – May 2024
Centre for Health Economics and Policy Innovation, host: Prof. Marisa Miraldo

Research Master in Economics, ECARES, Université Libre de Bruxelles — 2023
Supervisor: Mathias Dewatripont

M.Sc. in Quantitative Economics (QEM), Paris 1 Panthéon-Sorbonne and UCLouvain — 2022
Supervisor: Hervé Moulin

B.Sc. in Mathematical Statistics, School of Economics, Xiamen University — 2018

RESEARCH FIELDS

Primary: Economics of Innovation, Industrial Organization, Health Economics
Secondary: Applied Microeconomic Theory, Applied Econometrics

JOB MARKET PAPER

"Asymmetric Intellectual Property Rights and Follow-On Innovation: Evidence from Drug Repurposing"

Upstream patents are traditionally viewed as "gate" taxing follow-on innovation, as downstream innovators must negotiate licenses with transaction costs. This assumes well-delineated property rights provide sufficient incentives for both upstream and downstream innovators. However, when downstream property rights are poorly delineated, innovators capture limited economic rents and may instead leverage upstream patents to secure returns, turning them into a "shield."

Developing new uses for existing drugs exemplifies this, as secondary patents are often difficult to enforce. Instrumenting drug's market exclusivity with FDA review duration for new molecular entities (1985–2024), I find that an additional year of exclusivity increases expected post-approval new uses by 34%, with the marginal effect collapsing to zero at 15 years. Both drug owners and new entrants respond positively. These findings support the "shield" view and highlight a fundamental failure in delineating secondary property rights for drug's new uses.

PUBLICATIONS

S. Mermelstein, A. Gentilini, C. Yue, **X. Niu**, M. Shaikh, M. Dewatripont, H. Stevens, and M. Miraldo. "Do Policy Incentives Align with Pharmaceutical Innovation Challenges? A Retrospective Analysis of 42 Countries, 2000–2022." *Health Economics*, 2027 (invited submission, Special Issue).

S. Akodad, **X. Niu**, B. Secades, and H. Stevens. "Impact of Drug Repurposing between 1985 and 2024 on Pharmaceutical Innovation." *Communications Medicine* (Nature Portfolio), 6:84, 2026. doi:10.1038/s43856-025-01344-1

RESEARCH PROJECTS

Pharmaceutical innovation policy in 42 countries, Core Contributor — Brussels · 2023 – 2026
Hi-Prix Work Package 7, co-led by ULB and Imperial College London (Horizon Europe, €5.3M, 18 partner organizations in 10 countries); doctoral researcher funded by the project
Output: "Incentives for Pharmaceutical Innovation and Equitable Access to Innovation." Policy report for Hi-Prix, 2026. With M. Miraldo, H. Stevens, C. Yue, S. Mermelstein, A. Gentilini, M. Shaikh, and M. Dewatripont.

WORKING PAPERS

"Unbundling Global Innovation Policies: Mechanism Effects, Spillovers, and the Equitable Allocation of Pharmaceutical R&D." With C. Yue, S. Mermelstein, M. Miraldo, H. Stevens, M. Shaikh, and M. Dewatripont.

Innovation policies bundle several incentive mechanisms with offsetting effects, so estimated policy-level effects identify no single margin. Decomposing 1,100 policy introductions into six mechanisms and linking them to 1.9 million R&D grants across 42 countries (2000–2021), we exploit within-policy variation in mechanism composition to separate own-country effects, cross-country spillovers, and allocation relative to disease burden. Revenue subsidies raise R&D but redirect it toward commercially lucrative diseases and high-income markets, widening the gap between private and social returns. Mechanisms that lower regulatory and information frictions yield larger and more persistent gains, concentrated where profit incentives are weakest, and generate positive spillovers across borders. Mandated IP-sharing lowers appropriability and depresses short-run entry. Success-contingent rewards dominate unconditional transfers, consistent with reduced moral hazard and better selection.

WORK IN PROGRESS

"Patent Box Tax Relief and Orphan Drug Development."

Nine EU countries have extended preferential IP-box tax rates to orphan-drug income since 2015. Using difference-in-differences on a firm–country–year panel of 78,060 observations across 30 EMA member states (2000–2024), I find that adoption raises orphan-drug launches by roughly 75 percent. The effect is confined to large firms, with no response from SMEs and no increase in designations or registrations, indicating that the policy stimulates commercialization rather than innovation.

RESEARCH EXPERIENCE

Research Intern, Fair Division project (microeconomic theory), Paris 1 Panthéon-Sorbonne 2021 – 2022
Supervisor: Prof. Hervé Moulin

MENTORING

Master's thesis co-supervisor, with Hilde Stevens (I3h, Université Libre de Bruxelles)
Students: Alessandro Giarrizzo, Nour Hammoudi

PRESENTATIONS

2026–27 European Winter Meeting of the Econometric Society (EWMES), Warsaw (scheduled)
2024–25 Hi-Prix Consortium Annual Conference, Brussels (poster)
2023–24 Hi-Prix Consortium Annual Conference, Lisbon

FELLOWSHIPS AND AWARDS

Ph.D. Fellowship, Horizon Europe Hi-Prix project 2023 – 2026

SKILLS AND LANGUAGES

Theory: Industrial-organization models of cumulative innovation, patent breadth, and licensing.
Empirical methods: Panel data and causal inference (difference-in-differences, IV including shift-share, regression discontinuity, synthetic control); LASSO and random forest; questionnaire design and expert interviews.
Programming: Python, R, Stata, Quarto / R Markdown, LaTeX (proficient); SQL, Git (basics).
Data engineering: API-based collection (Dimensions, Cortellis); web scraping; PDF and LLM-assisted text extraction; record linkage for datasets with millions of records.
Domain data: FDA Orange Book and Patent Term Extension records; EMA orphan-drug designations; Cortellis Competitive Intelligence; Dimensions (publications, patents, grants, clinical trials); ICD-11; cross-country R&D tax, IP-box and HTA policy.
Languages: Chinese (native), English (fluent), French (basic).

REFERENCES

Micael Castanheira, FNRS Research Director and Professor, ECARES, ULB micael.castanheira@ulb.be
Mathias Dewatripont, Professor of Economics, ECARES, ULB mathias.dewatripont@ulb.be
Mattia Nardotto, Associate Professor of Economics, ECARES, ULB mattia.nardotto@ulb.be

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